Extreme Learning Machine for SOC Estimation of Lithium-ion battery Using Gravitational Search Algorithm

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Abstract— **This paper develops a state of charge (SOC) estimation model for a lithium-ion battery using an improved extreme learning machine (ELM) algorithm. ELM is suitable for SOC estimation since ELM algorithm has fast estimation speed, good generalization performance, and high accuracy. However, the performance of ELM is highly dependent on training accuracy and number of neurons in a hidden layer. Hence, gravitational search algorithm (GSA) is applied to improve the ELM computational intelligence by searching for the optimal value hidden layer neurons. The optimal ELM based GSA model does not require internal battery knowledge and mathematical model for SOC estimation. The model robustness is validated at different temperatures using different EV drive cycles. The performance of ELM-GSA model is verified with two popular neural network methods; back-propagation neural network (BPNN) and radial basis function neural network (RBFNN). The results are evaluated using different error rates and computation cost. The results demonstrate that the ELM-GSA model offers high accuracy and low SOC error rate than BPNN-GSA and RBFNN-GSA models. Furthermore, a detailed comparative study between the proposed model and existing SOC strategies is conducted which also demonstrates the superiority of the proposed model.**

Index Terms— State of charge, lithium-ion NMC battery, electric vehicle, extreme learning machine, a gravitational search algorithm

I. INTRODUCTION

Global emissions and climate change problems are a serious threat to human life. In order to reduce the carbon emissions, many developed countries around the world have already announced to substitute diesel and petrol based vehicle with electric vehicle (EV) in near future [1]. However, the EV is still struggling to become a popular and efficient mode of transportation due to the short travel distance and limited lifecycles of energy storage devices [2]. The performance of EV depends on reliability, safety, driving range, power management system and most importantly is highly dependent on energy storage functionality to predict and control critical health issues. Hence, the research on energy storage technologies in electric vehicles (EVs) needs intensive studies to improve the performance [3]. Therefore, further development is required to increase the storage capacity, minimize the catastrophic failures and incorporate a substantial balance to meet the operational requirements.

The lithium-ion battery has superior features such as long lifespan, high voltage, and energy capacity, fast-changing and low memory effect characteristics that outperform other energy storage devices [4]. The prognostic model development for lithium-ion battery with the estimation of the state of charge (SOC), state of health (SOH) and remaining useful life prediction has been researched thoroughly in recent days. Among the various components of battery management system (BMS), SOC is a vital parameter of the lithium-ion battery which indicates how much charge is available a battery cell. SOC delivers information about the driving mileage of EV. SOC is significant to control the overcharging and over discharging of lithium-ion cells. However, there are internal and external challenges, which cause difficulties to develop an accurate SOC estimation model. The internal challenges relate to the lithium-ion battery material, self-discharge, aging, thermal runway and hysteresis whereas the external issues are the ambient temperature variation and charging approach. The lithium-ion battery is very sophisticated to the said challenges. Therefore, further research on the SOC estimation model is required to extend the battery lifecycles as well as improve the EV performance. The mathematical expression of SOC is shown in (1) [5].

$$SOC = SOC_0 - \frac{1}{C_n}\int i.\eta dt \tag{1}$$

Where, SOC_0 and SOC represent the initial value and estimated value, respectively, C_n denotes the nominal capacity; t is the duration of battery discharging and η denotes the coulombic efficiency.

Different SOC estimation approaches have been highlighted in many literature. However, each approach has some drawbacks. The coulomb counting is the easiest method which can estimate SOC using current integration [6]. Nevertheless, coulomb counting method has cumulative effect problems [7]. Open circuit voltage (OCV) holds a relationship with SOC. Nonetheless, OCV method is time-consuming that is required for reaching a stable condition. Kalman filter is a well-known method for SOC estimation due to its capability to reduce high fluctuation and noises in current and voltage measurements [8]. Nevertheless, Kalman filter depends on battery model and mathematical relationship which may not be appropriate for highly nonlinear systems. Fuzzy logic and artificial neural network (ANN) use computational intelligence to estimate SOC with noise, aging and temperature effects [9]. Nonetheless, both methods need large memory device and an expensive processor for implementation.

To overcome the above problems, an improved algorithm based on optimization method is proposed under EV load

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profile with temperature effects. The ANN is useful and effective in SOC estimation since ANN has a strong intelligent algorithm that can predict battery non-linear dynamics without the need of any mathematical and battery model. ANN has the efficient computational capability that can deliver accurate results in changing conditions such as battery aging and ambient temperature. Several ANNs based SOC estimation have been explored such as radial basis function neural network (RBFNN) [10] back-propagation neural network (BPNN) [11] and. However, the parameter selection in the said ANNs methods were performed using ineffective trial and error method which was inefficient and wasted a lot of time and human energy. Furthermore, the algorithms suffered from data under fitting, overfitting problems, and slow computational cost. To address the above challenges, this study has proposed an extreme learning machine (ELM) which has good learning skill, high accuracy and fast response in training execution [12]. Nonetheless, ELM algorithm needs to find out the best value of neurons in the hidden layer to achieve high accuracy in SOC estimation. Hence, gravitational search algorithm (GSA) is employed to improve the computational intelligence of ELM algorithm. Lu et al. [13] estimated SOC using ELM method for a lithium-ion battery. Nonetheless, the hidden neurons were determined randomly which was not reasonable for obtaining good results. Therefore, this research has developed an improved SOC estimation model using ELM based GSA by selecting appropriate activation function, training algorithm, and a number of neurons to obtain high accuracy of SOC estimation.

II. Extreme Learning Machine

ELM has the efficient computational capability for predicting outcomes in complex and nonlinear systems. These unique characteristics of GSA are very useful for estimating SOC with high accuracy. Usually, one hidden layer is used for ELM learning. The parameters of the ELM need to be adjusted for improving the learning performance. The steps of ELM are described as follows.

i. At First, assign the parameters randomly. The input weight vector and hidden layer bias are represented as $a_i = [a_{i1,} a_{i2,} \dots \dots a_{iN,}]^T$ and b_i respectively where i is the number of neurons in hidden layer. The hiddea a n neurons are assigned as L. The value of L can be changed in order to achieve reasonable accuracy.

ii. Calculate the output matrix of the hidden layer. The mathematical expression is represented by

$$\sum_{i=1}^{L} B_i G(a_{i,}\ b_{i,}\ x_{j,}) = t_j, j = 1, \dots . N \quad (2)$$

Where B_i represents the output weight which connect the *ith* hidden layer neuron and output layer neuron. *G(x)* activation function which is determined before training. In this research, the most popular sigmoid function is used for the activation function [14].

$$G(a_{i,}\ b_{i,}\ x_{j,}) = \left(1 + e^{-(a_{i,} b_{i,} x_{j,})}\right)^{-1}, i = 1, \dots . , L, j = 1, \dots . , N \quad (3)$$

Equation (4) can be represented compactly as,

$$H\beta = T \quad (4)$$

Where

$$H = \begin{bmatrix} G(a_{1,}\ b_{1,}\ x_1) & \dots & G(a_{L,}\ b_{L,}\ x_1) \\ \dots & \dots & \dots \\ G(a_{1,}\ b_{1,}\ x_{N,}) & \dots & G(a_{L,}\ b_{L,}\ x_{N,}) \end{bmatrix}_{N \times L}$$

$\beta = [\beta_1, \dots . , \beta_L]^T$ and $T = [t_1, \dots . , t_N]^T$

H is the matrix of the output layer of the neural network.

iii. Calculate the output weight which is estimated by the inverse process of the hidden layer matrix [15] [16]. Hence, β is achieved by

$$\beta = H^+ T \quad (5)$$

where H^+ is the Moore–Penrose generalized inverse of H [17]

It is apparent from the equations mentioned above, ELM does not require any iterative method which results in substantial development in training speed. In summary, ELM has improved learning performance which gives better estimation results with high accuracy than other ANN algorithms.

III. Gravitational Search Algorithm

Rashedi et al. invented the GSA method in 2009 to achieve an optimal solution in any complex systems [18] . Rashedi et al. applied the concept of physics-based algorithms such as the law of gravity and mass interactions to develop GSA. GSA is based on the law of Newtonian gravity and laws of motion. The principle of GSA states that "every particle in the universe attracts every other particle with a force that is directly proportional to their masses and inversely proportional to the square of the distance between them" as expressed in the following equation [18].

$$F = G\frac{M_1 M_2}{R^2} \quad (6)$$

Where F denotes the magnitude of the gravitational force; G represents the gravitational constant; M_1 and M_2 characterize the mass of the first and second particles, respectively; and R is the distance between the two particles.

Newton's second law states a relationship between acceleration, force, and mass of particle which is expressed as follows;

$$a = \frac{F}{m} \quad (7)$$

A new term named Gravitational constant, $G(t)$ was introduced which is assessed using the initial value of the gravitational constant, $G(t_0)$ and the ratio of initial time t_0 and actual time t as follows;

$$G(t) = G(t_0) \times \left(\frac{t_0}{t}\right), \beta<1 \quad (8)$$

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Fig. 1 shows how the influence of mass forces between the particles develop the net force and acceleration.

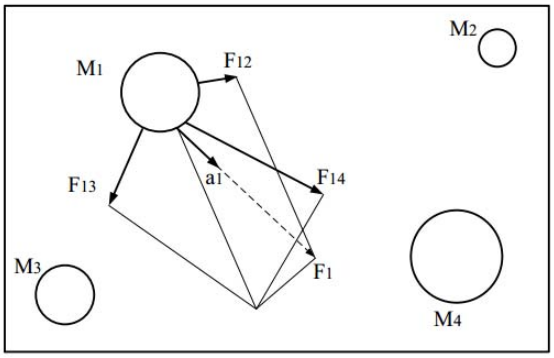


Fig. 1 Mass effects with other masses

The positions of the N number of the agents are initialized shown as follows:

$$X_i = \left(X_i^1, \ldots \ldots X_i^d, \ldots \ldots X_i^n\right), \quad for\ i = 1,2, \ldots, N \tag{9}$$

where X_{di} is the position of *i-th* agent in the *d-th* dimension and n is the space dimension. The mathematical equations for best and worst value and the masses of each agent, are presented as,

$$best(t) = min fit_j(t) \tag{10}$$

$$Worst(t) = max fit_j(t) \tag{11}$$

$$m_i(t) = \frac{fit_i(t) - Worst\ (t)}{best\ (t) - Worst\ (t)} \tag{12}$$

$$M_i(t) = \frac{m_i(t)}{\sum_{j=1}^{N} m_i(t)} \tag{13}$$

The total force F for the *i-th* agent is evaluated based on the gravitational constant G, position X, acceleration a. Then, the velocity and position are updated.

$$G(t) = G_0\, e^{(-\alpha t/T)} \tag{14}$$

$$F_{ij}^d(t) = G(t)\frac{M_{pi} \times M_{\alpha j}}{R_{ij} + \varepsilon}\left(X_j^d(t) - X_i^d(t)\right) \tag{15}$$

$$F_i^d(t) = \sum_{j \in Kbest,\, j \neq i} rand_j\, F_{ij}^d(t) \tag{16}$$

$$a_i^d(t) = \frac{F_i^d(t)}{M_i(t)} \tag{17}$$

$$v_i^d(t+1) = rand_i \times v_i^d(t) + a_i^d(t) \tag{18}$$

$$x_i^d(t+1) = x_i^d(t) + v_i^d(t+1) \tag{19}$$

GSA is dominant to another optimization algorithm in terms of fast convergences speed, however, GSA has the tendency to trap in local minima that limits its approach to expand the algorithm's population [19]. The steps of GSA pseudo code GSA are described as follows.

1. Search space identification, t=0
2. Randomized initialization X_i(t) for i=1,2,…., N
3. Evaluation of fitness function of agents.
4. Update the values of G(t), best (t), worst (t), $M_i(t)$ for i=1,2,…., N
5. Calculation of acceleration and velocity of agents.
6. Revise the agent position to yield X_i(t+1) for i=1,2,…., N, t=t+1
7. Repeat steps from 3 to 7 until maximum iteration is achieved.

V. Modeling Methodology

Firstly, SOC estimation for lithium-ion battery begins with the establishment of a battery test bench followed by data collection. Beijing Dynamic Stress Test (BJDST) and US06 drive cycle are used in this research. SOC has a high correlation with current, voltage and temperatures, hence these variables are selected for developing input dataset. The consecutive drive cycles are chosen since a higher number of data is required for data training. In order to validate the model adaptability and robustness, the data is recorded in two different temperatures (25°C and 45°C). The data normalization is executed before the training process is started. The entire dataset of BJDST and US06 cycles are categorized into two groups; training and testing. About 70% selected data is employed for model training and 30% selected data is used for data testing. The overall methodological framework of ELM based GSA for SOC estimation is illustrated in Fig. 2. The flowchart has three phases. Phase 1 describes the data collection method. Phase 2 covers the GSA execution process. Phase 3 highlights the SOC estimation and validation process.

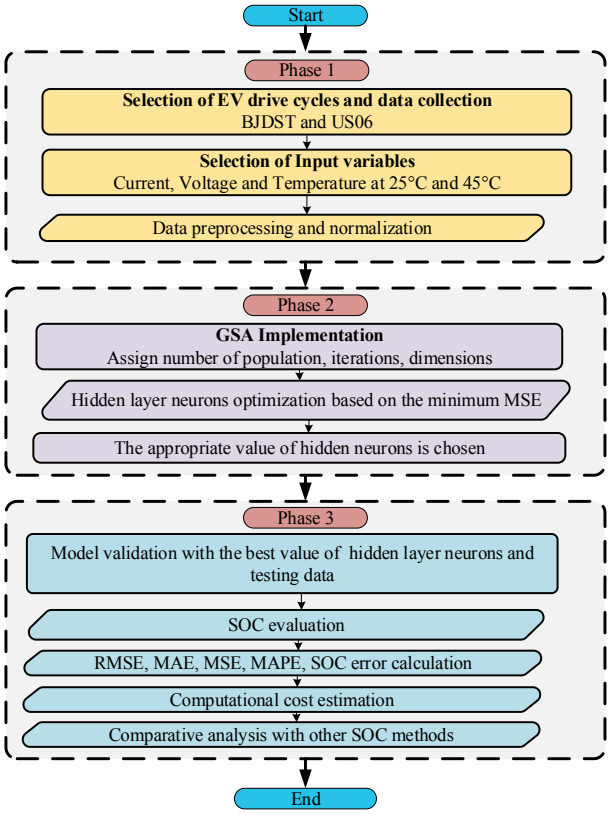


Fig. 2. Flowchart of ELM based GSA model

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ELM training process is combined with GSA in order to find the appropriate number of hidden layer neurons. The lowest fitness function determines the best value of hidden layer neurons. The combination of ELM and GSA enhances the accurateness and efficiency of SOC which confirms the best data fitting for the training function and delivers results with less error and less variance. Finally, SOC is assessed with testing data and ELM activation function together with an optimal number of neurons achieved form GSA. After SOC estimation, the proposed model is validated using computation cost and error rates. Finally, a comprehensive comparative study between the proposed ELM-GSA model and popular SOC estimation methods and is analyzed.

Phase I. Phase I covers the experimental arrangement, input variables selection and data collection at different temperatures. Three significant factors including voltage, current, and temperature are chosen. The following steps are conducted in Phase I.

i. A battery test bench was established using a battery cycler (Arbin BT2000), lithium-ion battery, thermal chamber and host computer as presented in Fig. 3. The experiment was conducted using 18650 lithium nickel manganese cobalt oxide ($LiNiMnCoO_2$ or NMC). Arbin BT2000 was employed to protect the NMC cell from being overcharged and undercharged. The thermal chamber was used to change the temperature as well as to monitor and record the temperature of NMC cell. All the data variations were observed in the one-second interval and controlled by the host computer. The applicability of the proposed model in real time EV application is verified using BJDST [20] US06 [21]. The driving time to complete one BJDST and US06 cycles are 916 seconds and 600 seconds respectively [20]. Battery discharge and data extraction were conducted using 12 consecutive BJDST and US06 cycles.

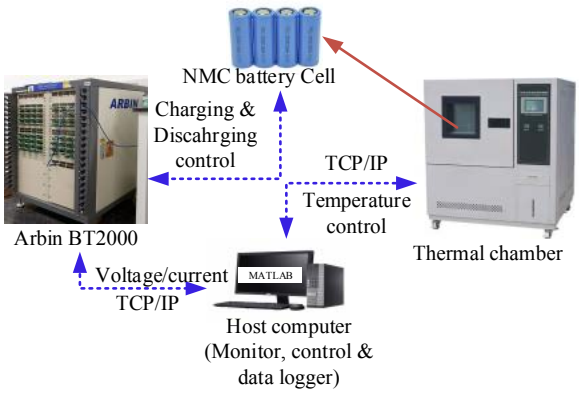


Fig. 3 Schematic diagram of battery test bench

ii. BJDST and US06 cycles are loaded to extract features at two different temperatures (25°C and 45°C).

iii. Data normalization is important in ELM training since it accelerates the training speed, and helps the algorithm to become more efficient and robust. The normalization data process is expressed in (21) where the length is set to [-1,1]

$$x = \frac{2\left(x - x_{\min}\right)}{x_{\max} - x_{\min}} - 1 \qquad (20)$$

Where, the minimum and maximum value of the input variable x of the ELM model is denoted by $x_{\max}$ and $x_{\min}$. The training and testing dataset are scaled using the same normalization range. The input dataset of BJDST and US06 drive cycles are shown in Fig. 4 and Fig. 5.

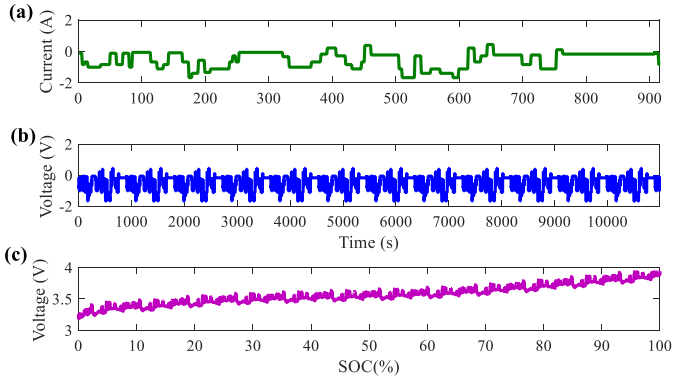


Fig 4. BJDST drive cycle input dataset (a) current in one cycle (b) voltage (c) voltage against SOC

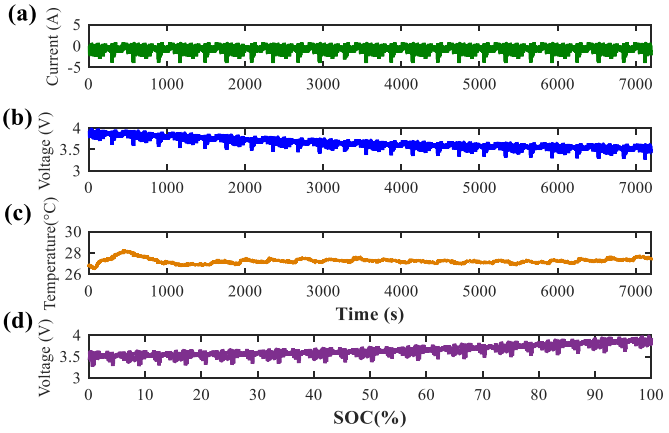


Fig 4. US06 rive cycle input dataset (a) current (b) voltage (c) temperature and (d) voltage against SOC

Phase II. GSA is utilized to determine the optimal number of hidden layer neurons. The following steps are used in phase II.

i. Firstly, the implementation of GSA algorithm starts with the setting of parameters. In this research, the population size (agent) and iteration number are assigned as 50 and 100, respectively. The study also defines the limit of hidden layer neurons. The boundary is between '0' and '20'.

ii. The agent position of hidden layer neurons is created in random order that are placed within the boundary range.

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iii. ELM training algorithm and activation function are executed to train the data of hidden layer neurons.
iv. The fitness function of each agent is evaluated. In this research, MSE is chosen as the fitness function.
v. The gravitational constant G, and best, worst position of the agent are updated using (14)-(16).
vi. Calculate the velocity and acceleration of agent using (18) and (17)
vii. Update agent position using (19).
viii. The training algorithm and activation function are recalled to re-evaluate the fitness value of each agent
ix. The best agent location and velocity are assessed through all the iterations based on the lowest value of fitness function. The minimum value of fitness function corresponds to the best value of hidden layer neurons
x. At the final step, SOC is estimated with the appropriate number of hidden layer neurons.

Phase III. Phase III is called the final stage where SOC is estimated with the testing data and the optimal value of neurons achieved from GSA. The accuracy of the ELM based GSA model is examined by various error terms. The summary of this phase is explained as follows.

i. The ELM algorithm takes the optimal value of hidden neurons and initiates training process.
ii. The testing data is used to monitor SOC under 25°C and 45°C temperatures. SOC is estimated in both BJDST and US06 drive cycles. The estimated SOC is checked with the reference value achieved from the coulomb counting method. In this study, the correct sensor was calibrated carefully to avoid cumulative error. Additionally, SOC error is observed which is calculated by the difference between the estimated SOC and reference.
iii. The effectiveness of the proposed model is evaluated using various types of error rate. The error rates include root mean square error (RMSE), MSE, mean absolute error (MAE) and mean absolute percentage error (MAPE). The equations of these error rates are presented as follows:

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(I_{es} - I_a)^2} \quad (21)$$

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(I_{es} - I_a)^2 \quad (22)$$

$$MAE = \frac{1}{N}\sum_{i=1}^{N}(I_{es} - I_a) \quad (23)$$

$$MAPE = \frac{1}{N}\sum_{i=1}^{N}\left|\frac{I_a - I_{es}}{I_a}\right| \quad (24)$$

Where I_{es} and I_a denote the estimated value, and actual value respectively and N denote number of observation.

VI. Results and Discussions

A. Fitness Value and Optimal Parameter

The fitness value of GSA is assessed by generating optimization response curves for BJDST and US06 cycles with 100 iterations, as illustrated in Fig. 6 and Fig. 7, respectively. The optimization response curves are observed at 25°C and 45°C temperatures. The specific iteration number which has the lowest value of fitness value delivers the best value of hidden neurons. In BJDST cycle, the lowest value of fitness value is recorded after 20 and 27 iterations. The values correspond to the said iterations are 2.3×10^{-3} and 1.9×10^{-3} at 25°C and 45°C respectively. On the other hand, 38 and 67 iterations provide the lowest value of fitness value amounting to 2.7×10^{-3} and 2.2×10^{-3} at 25°C and 45°C respectively in the US06 cycle. It is evident that the fitness value declines as the temperature rises. The electrolyte activity increases with the rise in temperature. Hence, the capacity of the battery rises as the temperate accelerates.

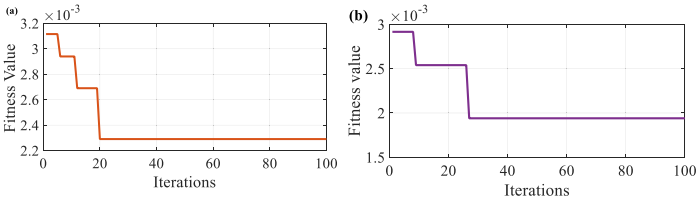


Fig. 6. optimization response curves of BJDST cycle (a) 25°C, (b) 45°

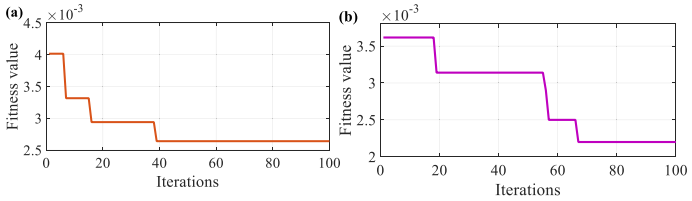


Fig. 7. optimization response curves of the US06 cycle (a) 25°C, (b) 45°

The optimal value of hidden neurons of 220, 328 are obtained in ELM based GSA method after 20 and 27 iterations at 25°C, and 45°C respectively in BJDST cycle. Similarly, in the US06 cycle, the optimal number of hidden neuron values of 158 and 147 are computed at 25°C, and 45°C respectively.

B. SOC Evaluation

ELM based GSA model examines SOC in BJDST and US06 cycle, as shown in Fig. 8 and Fig. 9 respectively. The SOC estimation results and SOC error rates are compared with the reference. Ampere-hour method with a calibrated current sensor is chosen as reference. Before loading EV drive cycles, NMC was fully charged and the initial SOC was set to 100%. It is seen for the figures that the estimated SOC value is almost placed to the same line of the reference value throughout the drive cycles which demonstrate that the proposed model has high robustness and low estimation error. It is also important to monitor SOC error which signifies how much estimated SOC deviates from the reference value. The error lies at between -3.1 and 3.6% and -5.6% and 6.5% for BJDST and US06 cycles respectively at 25°C. The estimation error in US06 is higher than that of BJDST due to the high

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randomness and fluctuation of load current. It is also observed from the figures that the SOC error declines with the rise in temperature. The error bounds are from -2.9% and 3% in BJDST whereas in the US06 cycle, the error bounds are from -4.8% to 5.3% At 45°C. The results achieved through simulation demonstrate that the proposed ELM based GSA model is a robust and efficient SOC estimation method that can perform under different operating conditions.

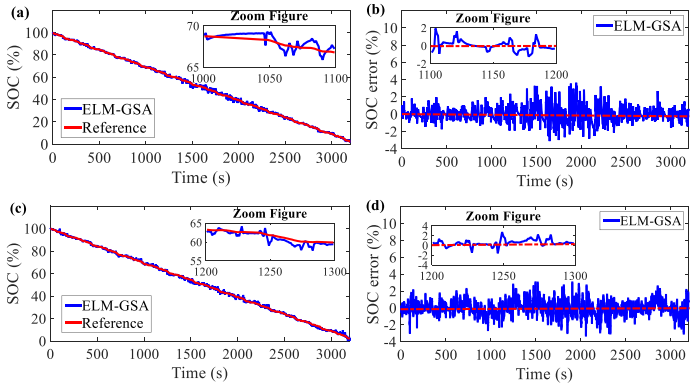


Fig. 8. (a-b) SOC (%) and SOC error (%) at 25°C, (c-d) SOC (%) and SOC error (%) at 45°C in BJDST cycle

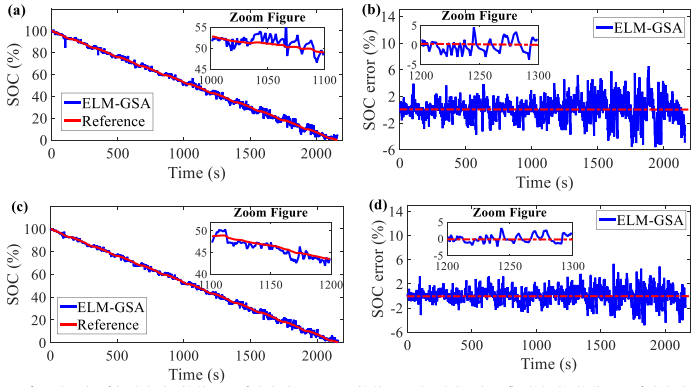


Fig. 9. (a-b) SOC (%) and SOC error (%) at 25°C, (c-d) SOC (%) and SOC error (%) at 45°C in US06 cycle

C. Computational Cost

This study also calculates the computation cost in order to validate the suitability of the proposed model in EV operation. ELM has a fast response to data training whereas BPNN and RBFNN models need a long time in training for weight and bias upgradation. However, higher number neurons make the ELM training process extended. It is significant to select the appropriate value of neurons, which will not only improve the estimation results but also increase the convergence speed. Therefore, this paper applies GSA inside an ELM network to find the suitable value of neurons. With the implementation of GSA, the convergence speed and SOC estimation results have improved. The computational cost in two drive cycles is shown in Table I.

TABLE I
COMPUTATIONAL COST IN BJDST AND US06 DRIVE CYCLE

Model	BJDST		US06	
Temperature	25°C	45°C	25°C	45°C
Hidden neurons	220	328	158	147
Computational Cost (s)	320.48	476.28	279.43	235.08

D. Performance Comparison

A detailed comparative analysis of ELM, BPNN, and RBFNN is presented in Table II and Table III. In order to perform a fair comparison, BPNN and RBFNN methods use similar input dataset for training and testing. Furthermore, the hidden neurons of BPNN and RBFNN are tuned based on GSA with the same number of iteration and population size. In BJDST cycle, ELM based GSA achieves RMSE of 0.76% whereas BPNN-GSA and RBFNN-GSA models obtain RMSE of 0.95% and 1.12%, respectively at 25°C. However, the lower error rates and SOC error are observed as temperature increases from 25°C to 45°C. The proposed model computes RMSE to be 0.68% at 45°C which is reduced by 21.8% and 29.8% from the BPNN-GSA and RBFNN-GSA model, respectively. An improvement is also noticed in MSE, MAE, and MAPE in ELM-GSA model under different conditions. The MSE of the proposed model is estimated to be 0.0059% that is dropped by 32.2% and 46.4% from BPNN-GSA and RBFNN-GSA models, respectively at 25°C. Similarly, about 23.6% and 35.3% reduction from BPNN-GSA and RBFNN-GSA model, respectively at 25°C are noted in the proposed model while assessing MAE. In addition, MAPE in BPNN-GSA and RBFNN-GSA models become higher raised by 3.6%, 8.1%, in comparison to EML-GSA model at 25°C.

TABLE II
PERFORMANCE ASSESSMENT IN BJDST CYCLE

Model	BPNN-GSA		RBFNN-GSA		ELM-GSA	
Temperature	25°C	45°C	25°C	45°C	25°C	45°C
RMSE (%)	0.95	0.87	1.12	0.97	0.76	0.68
MSE (%)	0.0087	0.0072	0.011	0.0092	0.0059	0.0052
MAE (%)	0.72	0.64	0.85	0.71	0.55	0.48
MAPE (%)	3.98	3.75	4.15	3.82	3.84	3.24
SOC Error (%)	-5.2~5.1	-4.8~4.6	-6.6~6.8	-5.3~4.9	-3.1~3.6	-2.9~3

The estimation results of the proposed model are further evaluated using the US06 cycle. RMSE in BPNN-GSA and RBFNN-GSA model is measured to be 1.68 %, 2.2%, respectively which is a 7.7%, 41% rise from ELM-GSA model. Moreover, there is a drop of 10.7%, 25.1%, in MAE in the proposed model in comparison to the other two models, respectively at 25°C. Furthermore, ELM-GSA obtains a 14.3% and 23.8% reduction in MAPE in comparison to the other two models at 25°C.

TABLE III
PERFORMANCE ASSESSMENT IN US06 CYCLE

Model	BPNN-GSA		RBFNN-GSA		ELM-GSA	
Temperature	25°C	45°C	25°C	45°C	25°C	45°C
RMSE (%)	1.68	1.38	2.2	1.88	1.56	1.2
MSE (%)	0.032	0.018	0.089	0.064	0.024	0.014
MAE (%)	1.87	1.12	2.23	1.35	1.67	0.91
MAPE (%)	16.71	9.28	18.8	12.7	14.32	7.5
SOC Error (%)	-6.2~8.7	-5.1~5.7	-8.7~9.2	-6.3~6.8	-5.6~6.5	-4.8~5.3

VII. COMPARATIVE VALIDATION WITH OTHER METHODS

The accuracy and robustness of ELM-GSA method is further validated by investigating the SOC error terms between the developed model and other prominent methods. Seven recent and prominent studies including backtracking search algorithm (BSA), Kalman filter, fuzzy logic, neural network, deep learning and support vector machine are considered for comparative evaluation, as presented in Table IV. It is evident from Table IV that, ELM based GSA has better performance and robustness than other existing approaches under different

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operating conditions. For instance, SOC error in BPNN based backtracking search algorithm (BSA) model is estimated within -2.1% ~8.8% and -2.4%~9.8% in DST and FUDS cycle,

TABLE IV
COMPARATIVE PERFORMANCE ASSESSMENT OF SOC ESTIMATION

No.	Method	Ref.	Test battery	Validation Profile	Temperature	Error rate
1.	OCV	[22]	18650 cylindrical type 1.1 Ah lithium-ion cells ($LiFePO_4$)	Federal Urban Driving Schedule (FUDS)	0 °C to 50 °C at an interval of 10 °C.	RMSE < 5%
2.	Electrochemical Model (ECM)	[23]	3 Ah $LiFePO_4$ cell	Urban Dynamometer Driving Schedule (UDDS)	25°C	SOC error < 5% in different RC networks
3.	Kalman Filter (KF)	[24]	20 AH A123 cylindrical 26650 Li-ion batteries	US06 drive cycle	25°C (ambient temperature)	SOC error <4%
4.	H∞ filter	[25]	1.2 Ah lithium-ion battery	Pulse charging and discharging	Room temperature	RMSE under 2.5%
5.	BPNN based BSA	[26]	2 Ah 18650 $LiNiMnCoO_2$/NMC	(i) FUDS (ii) Dynamic Stress Test (DST)	0°C, 25°C and 45°C	RMSE 0.57%~1.74% in FUDS RMSE 0.48% ~1.47% in DST
6.	Deep Learning	[27]	2.9 Ah Panasonic $LiNiCoAlO_2$/ NCA	Dynamic drive cycles, ±18 A,	0°C, 10°C and 25°C	RMSE 1.11% ~2.44% MAE 0.77%~2.08%
7.	ANFIS	[28]	40 Ah, 36 Lithium-Ion cells	Charging and discharging currents (i) 57 A, (ii) 68 A	Temperature range from -30 °C to 55 °C	MSE 3.4% in 57 A MSE 5.6% in 68A
8.	ELM based GSA (proposed in this study)	-	2 Ah 18650 $LiNiMnCoO_2$/NMC	(i) BJDST (ii) US06	25°C and 45°C	RMSE 0.68%~0.76% in BJDST RMSE 1.2% ~1.56% in US06

respectively at varying temperatures. Nonetheless, the ELM based GSA model delivers more accurate SOC estimation results, with the SOC error being within -2.9%~3.6% and -4.8%~6.5% in BJDST and US06 cycle, respectively under varying temperatures. Similarly, ELM based GSA is dominant to deep learning method, fuzzy logic, Kalman filter, H∞ filter and SVM methods.

VII. CONCLUSION

An enhanced and robust estimation model based on ELM with GSA is suggested in this research for the estimation SOC estimation of lithium-ion battery. Generally, the hidden neurons of ELM are chosen randomly or by experience, which is not accurate and delivers poor estimation results. Hence, a recent optimization algorithm named GSA is employed to improve the ELM computational capability by finding the hidden neurons. The overall contribution is the development of an intelligent SOC estimation model using ELM with GSA that obtains improved performance in enhancing the robustness, improving the accuracy and accelerating the estimation speed under different EV drive cycles and temperatures. The significant outcomes of this research are summarized as follows: (1) satisfactory results are achieved with ELM based GSA model without the need of battery model and complicated numerical relations, (2) The robustness of the ELM-GSA model is verified using two EV drive profiles. The ELM-GSA algorithm improves SOC accuracy and obtains SOC error under 4% in BJDST cycle at 25°C (3) the ELM based GSA model achieves lower RMSE, MAE, MSE and MAPE than BPNN based GSA and RBFNN based GSA models (4) the convergence speed of ELM model is higher than BPNN and RFB models. The proposed model is further validated with other prominent SOC estimation techniques in the literature. The detailed comparative investigation indicates that the developed model is dominant to other approaches in obtaining high adaptability, efficiency, precision, robustness and estimation cost. The proposed model could be tested by developing a prototype embedded system in online EV application. Our future research is to estimate SOH for lithium-ion battery with aging effects.

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